

Supplementary document

Characterization: Exploring Materials Science

Explore essential methods and physical principles to determine the mechanical and electromagnetic properties of the material.

Using thermodynamics and radiation-matter interaction to evaluate performance and structure.

MECHANICAL TESTS AND RESISTANCE ELECTROMAGNETIC AND OPTICAL BASIS

Infrared Visible Light Ultraviolet
(IR) (500-770 nm) (UV)

The Optical Spectrum ><

Use visible light and
invisible radiation (IR/UV).

Traction Tests
and Compression

Determine Young module
and the yield strength by breakage.

Source
Radiation

MEB/MET

Electrons
Secondary
8 S-N Electrons s
Fatigue and Legtron Mechanics; Interaction
Eodite cycles Radiation-Material
Measuring cycle resistance Visualizes atomic structure by
and propagation of cracks. bombardment of electrons or photons.
N

HR

operating
5 ©

—— COMPARISON OF SPECTRAL GAMES

EEC XI

2 g to
Analysis of Hardness AS 70th 0th Poynting Energy Flow
Evaluates penetration resistance via Diffraction and Analysis Optical Observation and Thermal Analysis
and Describes radiation power

Rockwell or Vickers. internal structure reflection properties meecular vibrations traversing one surface per unit of time.

"M Gemini Notebook